1. **Scenario:** A program needs to determine the largest prime factor of a given number for mathematical computations.

   Write logic to find the largest prime factor of a given number.

   1• Divide the number by **2** repeatedly the modulo division (remainder) is not equal to zero..

   2• to handle factors 3 and upward; starting from **3** up to the square root of the remaining number

   # We only need to go up to the square root of n (ie) square of factor(iteration number) is not greater than given number

   3• Whenever the number is divisible by the current factor, divide the number and update the "largest factor" variable.

   4• If the remaining number is greater than 2, that number itself is a prime and thus the largest factor.

   Say for example take number 13195

   1 it is not divisible by 2

   ` 2 move on to 3 and iterate through odd number

   3 the number is divisible by 5 so take 5 as the max_prime and divide given number by5 we get 2639

   4 continue to iterate and do modulo division, when we reach 7 ➔2639 is divisible by 7 ; now replace max prime as 7 and update divide 2639 by 7 we get 377, take 377 for further processing

   5. on iterating further we see 13 can divide 377 which gives us remainder 29 since that number is greater than 2 and that cannot be divided further so now 29 become the max prime.

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2. **Scenario:** A system needs to convert a Roman numeral string into its integer equivalent for numerical processing.
   Write logic to convert a given Roman numeral string into an integer.

   1 Define a dictionary linking each symbol (I, V, X, L, C, D, M) to its value.

   2 **Iterate Right-to-Left:** It is often easier to start from the end of the string. Or reverse string before proceeding further to do process from left to right

   3 value checking**:**

   - If the current value is less than the value to its right, subtract it from the total.
   - Otherwise, add it to the total.

3. **Scenario:** A text-processing application analyzes a list of words to find the longest common prefix.
   Write logic to find the longest common prefix among a list of strings.

   1. Start with first word as the prefix,
   2. then compare it to the next word.
   3. shorten the prefix until it matches the start of the next word
   4. continue further on to next word, and repeat this for the entire list.

4. **Scenario:** A calculator program takes two numbers and an operator to perform a mathematical operation.
   Write logic to take two numbers and an operator (+, -, *, /) and perform the corresponding operation.

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1. Take input form user the first number 2nd number and the operand(+ - * /)
2. Based on the inputted character
3. If the operand is
    a. “+” : do summarion of both numbers
    b. “-“: subtract num2 from num1 and return the difference
    c. “*” : multiply both number and return the result
    d. “/” : divide num1 by num2 and return result

5. **Scenario:** A lottery system assigns special numbers where Armstrong numbers have a special significance.
   Write logic to check whether a given number is an Armstrong number.

   To see if it is a Armstrong number

   1.find total number of digits

   2. find value of each digit raised to the power of the total number of digits,

   3. total all the digits value raised to the power of total number of digits

   4. if the taotal equals the original number. Then that number is an Armstrong number

6. **Scenario:** A task management application allows users to add, remove, and view tasks in a to-do list.
   Write logic to implement a to-do list where users can add, remove, and view tasks.

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1. Use a Python list to store task strings.
2. To_Add: Use .append() to add a new string to the end of the list.
3. To_Remove: Use .pop() with an index or .remove() with a value to delete a task.
4.View: Use a for loop with to display tasks

7. **Scenario:** A weather application requires a function to convert temperature between Celsius and Fahrenheit.
   Write logic to convert a given temperature from Celsius to Fahrenheit and vice versa.

Celsius to Fahrenheit:

F = (C *(9/5)) + 32

Fahrenheit to Celsius:

C = (F - 32) * (5/9)

This is a straight forward implementation of above formula

8. **Scenario:** A security system needs to verify whether a given string has all unique characters.
   Write logic to check whether all characters in a given string are unique.

1.To check if all characters in a string are unique,
2. the most efficient approach is to compare the length
3. that of the original string with the length of a **Set** created from that string.

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4. Since sets in Python automatically remove duplicate elements, 5.any reduction in length proves that duplicates were present.

9. **Scenario:** A text analyzer tool identifies the longest word in a given sentence for readability analysis.

Write logic to find the longest word in a given sentence.

1. Split the sentence into a list of words, removing spaces.
2. Loop through each word in the list.
3. Check `len(word)` for each item.
4. Keep a variable that updates whenever a longer word is found
5. by the time all words are checked we will have the longest word will be identified..

10. **Scenario:** A password manager generates secure passwords based on user-defined length and complexity.

Write logic to generate a random password of a given length, containing uppercase, lowercase, digits, and special characters.

1Put together all required characters: uppercase letters, lowercase letters, digits, and special characters.

2 To guarantee the password meets complexity requirements, start by picking at least one character from each required category.

3. Fill the remaining length of the password by picking randomly from the entire pool of characters.

4. Shuffle the resulting list of characters to ensure we are not leaving behind any pattern in selecting special characters for complexity

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